

Empowered Action: An Infant Mortality Study
Capstone 2 Final Report

Landon Nguyen & Alex Nguyen

DATA 4382

Dr. Masoud Rostami

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Introduction

Infant mortality rate (IMR) is a crucial health metric that represents the general quality of a region's public health infrastructure. Nine states in India make up the Empowered Action Group; these states were identified as "lagging behind" in terms of public health and are the base problem that this study aims to understand. More specifically, the goal of this project was to create a model that can accurately predict the IMR for a given district and, therefore, determine the greatest districtwise factors contributing to poor public health; this could enable district-specific planning for improving public health and quality of life. Expanding on the baseline model and initial strategy from Capstone 1, the goal of this study was to improve model performance via stronger preprocessing and advanced modeling architectures. Furthermore, this study also aimed to utilize Explainable AI (XAI) to determine the greatest contributing factors to each district's predicted IMR and implement XAI into model deployment to create a tool capable of driving public health solutions.

Background

The Infant Mortality Rate (IMR) is widely regarded as one of the most sensitive indicators of a nation's social development and the effectiveness of its healthcare delivery systems. In the context of India, national averages often mask significant regional disparities. To address these inequities, the Government of India identified the Empowered Action Group (EAG), a collective nine states (including Bihar, Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, Rajasthan, Uttar Pradesh, and Uttarakhand) that have historically lagged in demographic transition and public health outcomes.

These regions face unique challenges, including varying levels of literacy, access to maternal care, and immunization coverage. While state-level policies provide a broad framework, public health experts emphasize that district-wise analysis is essential for identifying areas that are lacking. By understanding the specific socioeconomic and clinical drivers within these 284 districts, stakeholders can transition from generalized health mandates to targeted, evidence-based interventions. Data from the Annual Health Survey provides the necessary granularity to explore these variables, offering a path to reduce infant mortality rates through localized public health strategies.

Methodology

Data Understanding

The dataset used for this project was procured by the Annual Health Survey and shared via [Kaggle](#). Data was collected from 284 districts across 9 Empowered Action Group states in India: states identified as "lagging behind" with regards to public health. The dataset was tabular, consisting of 284 rows where each row represented a single district; there were 644 columns, including the target variable, that were grouped into 26 classifications indicated by a two-letter prefix (AA, BB, etc.). After acquiring

the data, the first stage of this project was exploring the dataset for any outstanding properties that could provide insight or interfere with modeling. The most notable features of the dataset included the target variable distribution, null value patterns, multicollinearity, and potential data leakage.

The target variable for this project was total infant mortality rate (IMR). The total IMR variable was distributed (see Fig. 1) to identify any potential quirks or outliers.

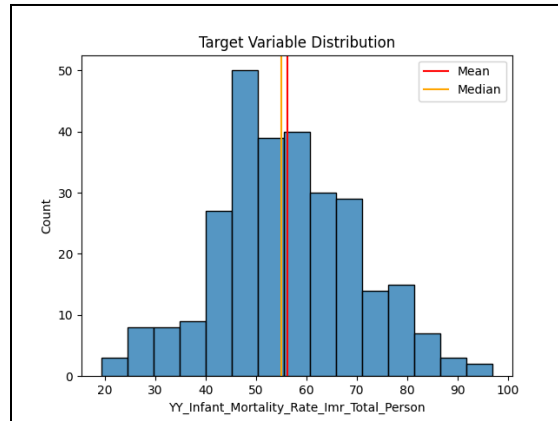


Figure 1. Distribution of the target variable (total IMR). The mean is marked with a red line, and the median is marked with a yellow line.

The distribution shows a mostly normal spread, which is further validated by the mean and median being nearly the same, 55 and 56, respectively. A lack of skewness implied that there should not be any notable outliers in the target distribution and, therefore, would not be a major issue for modeling.

Next, null values were inspected for any peculiar patterns so that a proper strategy could be developed to handle them. In total, 199 features contained null values. Interestingly, many features contained the same number of null values, so a missingness matrix was created to identify any potential patterns within these features (see Fig. 2).

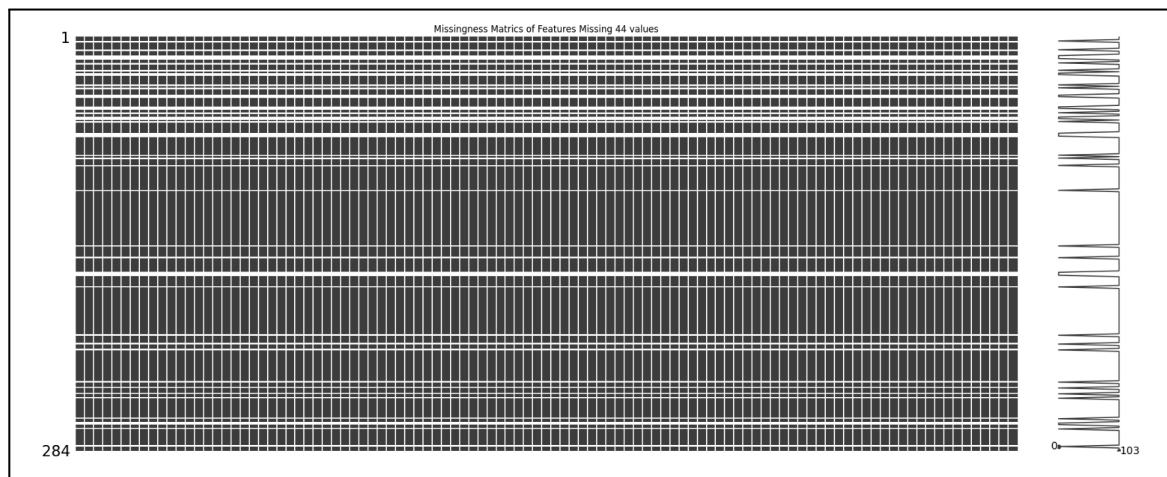


Figure 2. Missingness matrix for features containing 44 null values.

The matrix displays a uniform pattern for null values. That is, any feature with the same number of null values was missing information for the exact same districts. Here, a missingness matrix was shown for features missing 44 values, but this pattern also exists for features missing 45, 72, 98, etc. values.

Additionally, one of the most prominent problems for datasets with high dimensionality is multicollinearity. One unique property of this dataset was that features were already grouped together into major classes, indicated by their two-letter prefix. Under the assumption that high correlations would exist within these groups, heatmaps were created to inspect their interrelationships (see Fig. 3).

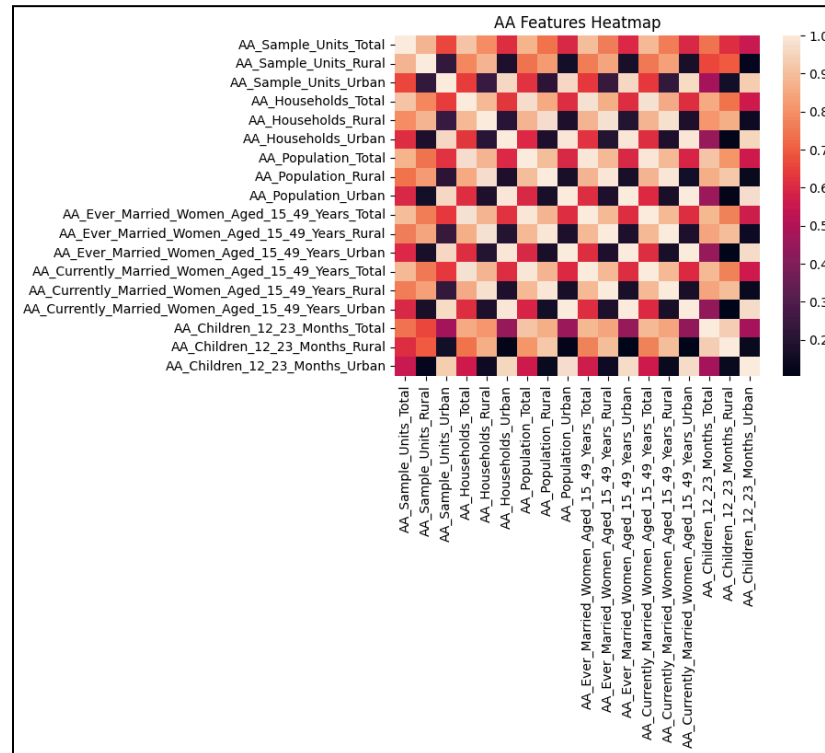


Figure 3. Heatmap of features belonging to the AA subgroup.

The heatmap shows numerous hot-spots where high correlation exists between the variables. As such, the subgroup relationships are used later in handling the multicollinearity problem.

Finally, one interesting property of the dataset was that the target variable, total IMR, belonged to a subgroup that contained other IMR values split by gender and urbanization as well as other mortality measurements. Therefore, a correlation table was created to inspect the relationships between these features and the target variable (see Fig. 4)

Correlations with Target Variable	
Variable	Correlation
YY_Infant_Mortality_Rate_Imr_Total_Person	1
YY_Infant_Mortality_Rate_Imr_Total_Male	0.97
YY_Infant_Mortality_Rate_Imr_Rural_Person	0.97
YY_Infant_Mortality_Rate_Imr_Total_Female	0.97
YY_Infant_Mortality_Rate_Imr_Rural_Male	0.95
YY_Neo_Natal_Mortality_Rate_Total	0.94
YY_Infant_Mortality_Rate_Imr_Rural_Female	0.93
YY_Under_Five_Mortality_Rate_U5MR_Total_Person	0.92
YY_Under_Five_Mortality_Rate_U5MR_Total_Male	0.91
YY_Neo_Natal_Mortality_Rate_Rural	0.91

Figure 4. A table of correlations between the target variable and other variables in its subgroups (YY). Figure 4 shows that some of the features in the same subgroup as the target variable had nearly perfect correlation with it. As such, these features posed a major data leakage problem.

Data Preprocessing

Before proper models could be created, the major quality issues in the dataset needed to be addressed. The first step taken to prepare the dataset for modeling was handling the data leakage problem. Data leakage could interfere with any step throughout the project and lead to misleading results. Therefore, features with more than 70% correlation with the target variable were removed. This included all of the other IMR features and numerous other mortality measurements.

After clearing the data leakage problem, null values were handled. The null values posed an interesting challenge because the limited number of rows and excessive number of features meant that dropping rows would not be a viable option. Furthermore, the uniform missingness pattern would make it difficult to impute values based on the features' relationships.. One notable relationship existed between the null values and the states. That is, the proportion of null values was different at the state level, and the distributions of each feature differed between the states. Therefore, missing values for features missing at most 44 values were imputed with the state median. Features with more than 44 missing values were removed from the dataset.

Following null value imputation and removal, the dataset was split into 70/30 train and test sets. A validation set was not used as there was too little data. A 70/30 split allocated ~200 observations for training which, with proper feature selection, should be enough to train a working model.

Next, the state names needed to be encoded and the features needed to be scaled to prevent large scale features from dominating the predictions. State name was target encoded to avoid adding additional dimensions to the dataset via one-hot encoding. Then, robust scaling was applied to the dataset. The dataset contained a mixture of skewed and normally distributed variables, making standard scaling non-optimal. Furthermore, the high dimensionality of the dataset makes handling skewness for individual

variables inefficient. Robust scaling, as the name implies, is robust to outliers; this makes it ideal for scaling a high dimensional dataset that contains a mixture of skewed and normally distributed variables.

Finally, multicollinearity was handled via a two part feature selection. In the first part, the featureset was split into 20 subsets to separate features with the same prefix (classification); doing this minimized the impact of multicollinearity on the feature selection process. Forward selection was then applied to each subgroup individually, and the top features from each group were aggregated; this left 50 remaining features from the original 643. In the second part, agglomerative clustering was applied to the remaining 50 features, based on the correlation matrix, which resulted in 14 clusters of related features (see Fig. 5).



Figure 5. The 50 features separated into 14 clusters as a result of the agglomerative clustering.

As seen in Figure 5, each cluster contains similar features and, therefore, can be assigned a fitting identity. Using PCA, each cluster was then compressed into one component, that is, each cluster became one feature in the final featureset.

Initial Modeling

Originally, the selected models included linear regression and Random Forest. Linear regression is a standard linear model that uses linear relationships to determine its predictions. Linear regression was chosen to test the strength of the dataset's linear relationships and act as a baseline model due to its simplicity. Random forest is an ensemble of decision trees that recursively partitions the dataset based on feature values to make a prediction. In opposition to linear regression, Random Forest was chosen to test the viability of nonlinear relationships and complex model architectures.

Baseline Linear Regression Metrics				Random Forest Metrics			
Subset	R2	Adjusted R2	RMSE	Subset	R2	Adjusted R2	RMSE
Train	0.83	0.81	5.67	Train	0.80	0.78	6.15
Test	0.83	0.80	5.86	Test	0.71	0.65	7.76

Figure 6. Comparison of the linear regression baseline and Random Forest.

The evaluation metrics used for this project include R^2 , adjusted R^2 , and root mean squared error (RMSE). R^2 represents the proportion of variance in the target variable explained by the predicted values, thus acting as a form of accuracy. Adjusted R^2 is similar but accounts for the dimensionality of the dataset. RMSE represents the average error of the model and is a standard metric for comparing model performances. Figure 6 displays two key insights. First, linear regression has a balanced fit (even train and test performance) but lacks the depth for stronger predictions. Second, Random Forest, a nonlinear model, makes stronger train predictions, but its complexity leads to overfitting.

Fine-Tuning

All tree-based models led to overfitting, so fine-tuning emphasized linear models. Numerous methods were attempted to boost the performance of the linear regression model, including L1 (Lasso) and L2 (Ridge) regularization. One notable performance boost came from adaptive boosting (AdaBoost) using a linear regression model as the based model, which slightly improved the test metrics. The most notable performance boost came from changing model architectures to a custom artificial neural network (ANN; see Fig. 7).

Layer Type	Input Shape	Output Shape	Param #
Sequential	[14]	[1]	4673
Linear	[14]	[64]	960
Dropout	[64]	[64]	0
Linear	[64]	[32]	2080
Linear	[32]	[32]	1056
Linear	[32]	[16]	528
ReLU	[16]	[16]	32
Linear	[16]	[1]	17

Figure 7. Custom artificial neural network architecture.

ANNs are inherently linear as the predictions come from a sum of slopes and biases, and nonlinearity is integrated via activation functions. The custom ANN was designed to have mostly linear layers with a single ReLU activation function at the end of the network; this added just enough complexity to surpass the linear models without overfitting like the tree-based models. Overfitting was accounted for in the ANN structure using a dropout layer, which forced the network to explore different pathways. This model trained for 10,000 epochs using mean squared error loss (MSE) and Adam optimization at an initial learning rate of 0.0005. A step scheduler was used to slow the learning rate to 0.0001 in the second half of

training; this slowed the model's learning to prevent it from overfitting. Both optimized models, AdaBoost and ANN, performed better than the baseline and Random Forest models (see Fig. 8).

AdaBoost Linear Regression Metrics				ANN Metrics			
Subset	R2	Adjusted R2	RMSE	Subset	R2	Adjusted R2	RMSE
Train	0.83	0.81	5.63	Train	0.86	0.85	5.10
Test	0.84	0.80	5.78	Test	0.87	0.85	5.10

Figure 8. Optimized AdaBoost and ANN metrics.

Figure 8 shows the metrics achieved by an adaptive boosted linear regression model and the custom ANN architecture. Both models achieved an adjusted R^2 at or above 0.80. The ANN model performed the best, being the first model to surpass an adjusted R^2 value of 0.85. Furthermore, The ANN was the first model to have balanced train and test metrics.

Explainable AI (XAI) Analysis

Model predictions were interpreted using SHapely Additive exPlanation (SHAP) values, starting with a global analysis (see Fig. 9), that is, the general impact each feature had on the model's predictions.

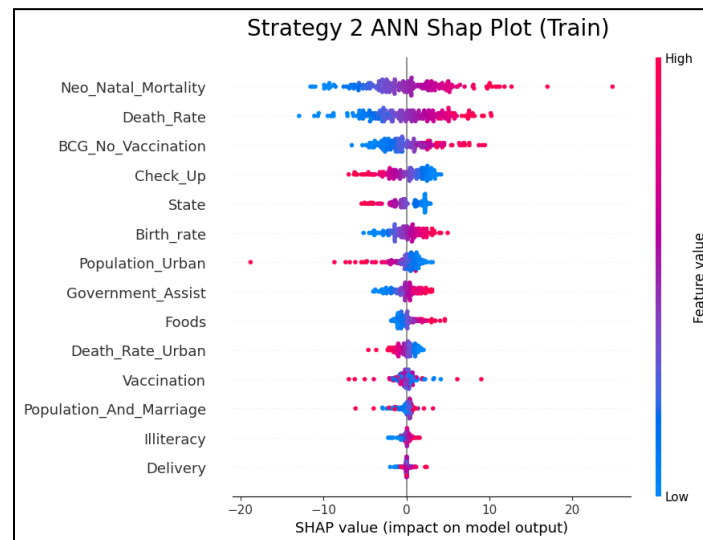


Figure 9. SHAP beeswarm plot for the ANN model.

The SHAP plot shows that the most impactful features were neonatal mortality, death rate, BCG vaccination coverage, and health check ups; these all align with modern research and expectations. The SHAP plot also shows how the model perceives the relationship between each feature and the target variable. Increases in death rate and neonatal mortality clearly increase the predicted IMR. Furthermore, a greater coverage lacking BCG vaccinations (BCG_No_Vaccination) is also a strong driver of increased IMR, and vice versa. Thus, the learned relationships for the top three features falls within expectation, which validates the models predictions. Some features, however, may require further inspection. For example, the mixed results for general vaccination coverage and the inverse relationship seen for check ups.

Model Deployment

The best performing ANN model was deployed via Streamlit. The deployment architecture consisted of a home page and two model selections: the full ANN model and a smaller ANN-mini model. The full ANN model is the best performing model from the pipeline. A user can input data (the 50 top features from the feature selection process and the district name) by either uploading a CSV file or manually inputting data by following the provided template. Users only need to ensure the data has no missing values; a preprocessor is built in to handle encoding, scaling, and PCA compression. The ANN-mini model works similarly, but only requires 14 feature inputs (and the district name) at the cost of some performance. The mini model also has a single-input option, which provides a separate input bar for each feature; this is ideal for faster predictions and rough estimates.

After inputting data into the model, both models follow a similar workflow. The data is preprocessed to fit the selected model → the model makes predictions for the inputs (regardless of the number of inputs) → the deployment app returns a table of the provided district names and the predicted IMR values. The user is then given a series of optional functions. The first function returns a SHAP waterfall plot to display how each feature contributed to the final prediction for a selected district. The second function uses an agentic AI to interpret the SHAP plot and describe how each feature relates to the predicted IMR to the user (see Fig. 10).

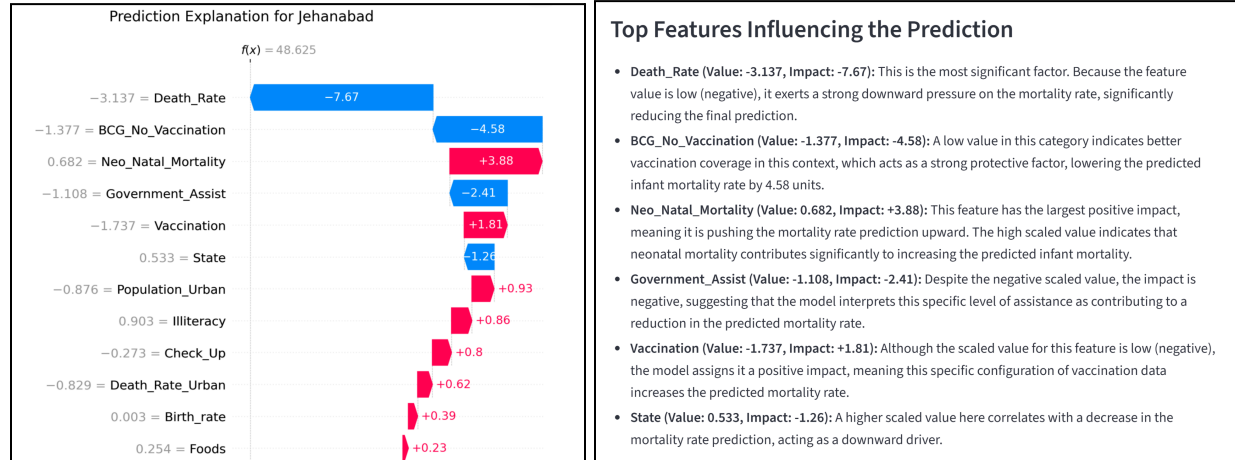


Figure 10. Example of the SHAP waterfall plot and agentic explanation produced by the deployment app. The last function uses a 3 agent system to combine the SHAP values with a web search to write a targeted report on how the features are impacting IMR for the specified region and recommend solutions for reducing IMR and improving public health (see Fig. 11).

2. Analysis of Key Drivers (SHAP Insights)

Through SHAP (SHapley Additive exPlanations) analysis, several high-impact variables have been identified:

- **Neonatal Mortality (Impact: +3.88):** Clinical complications such as sepsis, preterm birth, and asphyxia remain the most significant contributors to mortality.
- **BCG Vaccination (Impact: -4.58):** The lack of full immunization, particularly the BCG vaccine, serves as a major risk factor. Closing this gap offers the highest leverage for improving survival rates.
- **Government Assistance (Impact: -2.41):** Existing programs are underutilized. Strengthening the implementation of initiatives like Janani Suraksha Yojana is essential to reduce the burden of costs on families.
- **Maternal Education & Antenatal Care:** Low literacy rates correlate strongly with a lack of postnatal check-ups, exacerbating infant vulnerability.

3. Recommended Strategic Interventions

A. Clinical Infrastructure Expansion

Priority must be given to the upgrading of Sick Newborn Care Units (SNCUs) across the district. These facilities require functional resuscitation equipment and continuous, skilled clinical oversight to manage neonatal emergencies.

B. Immunization and Preventive Care

Closing the immunization gap is non-negotiable. Strategies should transition from passive service delivery to active, door-to-door tracking of infants to ensure 100% coverage, specifically targeting the BCG vaccine to leverage its significant protective impact.

C. Community-Based Outreach

Given the socio-economic barriers (illiteracy and rural isolation), ASHAs (Accredited Social Health Activists) must be empowered to bridge the gap in antenatal care, which currently remains low. This involves educating mothers on nutrition, birth spacing, and the importance of facility-based births.

Figure 11. Example of the agent report produced by the deployment app.

Figure 11 demonstrates the deployment agent's ability to interpret feature contributions to a given IMR prediction and use them as a basis for recommending public health improvements.

Challenges and Solutions

The first major challenge encountered was the dimensionality of the dataset. With 643 features and only 284 rows, any models were guaranteed to overfit. Furthermore, with this being a health analytics dataset, applying PCA and losing the features' definitions would pose ethical concerns. However, feature selection, such as forward or backward selection, would suffer due to the multicollinearity in such a large featureset. To circumvent these issues, a two stage feature selection process was implemented to reduce the dimensionality while working around the multicollinearity. In the first stage, by splitting the features into subgroups before forward selection, each subgroup could undergo selection without suffering from multicollinearity. In the second stage, clustering the remaining 50 features into related groups and applying PCA to each cluster further reduced the featureset to 14 components that could still be defined based on the features composing each cluster. Ultimately, this feature selection process enabled models to learn without severely overfitting.

The second major challenge was the limited number of observations. Linear models performed well, but were limited in depth. Additionally, more complex ensembles struggled with severe overfitting. ANN posed as an unexpected intermediate, especially since neural networks typically need a lot of data to perform well. However, ANNs are inherently linear, and a tested ANN architecture with only linear layers performed the same as a linear regression model. Adding a single ReLU activation function at the end of the structure, right before the output, gave the model enough nonlinear potential to surpass the linear regression models, but not so much complexity that the model would always overfit. Furthermore, including an early dropout layer right after the input integrated regularization to further prevent overfitting. Lastly, using a scheduler to slow the learning rate in the second half of the training loop and stopping the training loop early also helped prevent the model from overfitting. Ultimately, the custom ANN architecture and training structure resulted in a good-fit model that performed better than previous attempts while remaining balanced.

Key Results and Insights

Numerous models were produced, but the most important comparison to be made is between the baseline model, a tree-based model ensemble (Random Forest), an optimized linear model (AdaBoost), and the custom ANN model (see Fig. 12)

Model Comparison (Train Metrics)				Model Comparison (Test Metrics)			
Model	R2	Adjusted R2	RMSE	Model	R2	Adjusted R2	RMSE
Linear Regression Baseline	0.83	0.81	5.67	Linear Regression Baseline	0.83	0.80	5.86
Random Forest	0.80	0.78	6.15	Random Forest	0.71	0.65	7.76
AdaBoost Linear Model	0.83	0.81	5.63	AdaBoost Linear Model	0.84	0.80	5.78
ANN	0.86	0.85	5.10	ANN	0.87	0.85	5.10

Figure 12. Train and Test comparison tables of the most important models.

From the table, it can be concluded that nonlinear models perform significantly worse than linear models, which implies that most fluctuations in IMR can be explained through linear relationships. However, the best performance comes from the ANN model; this implies some level of complexity within the data's relationships.

The global SHAP analysis from before validated the model's predictions by justifying its most important learned relationships. However, Looking at the SHAP explanations of individual predictions reveals some potential inconsistencies (see Fig. 13).

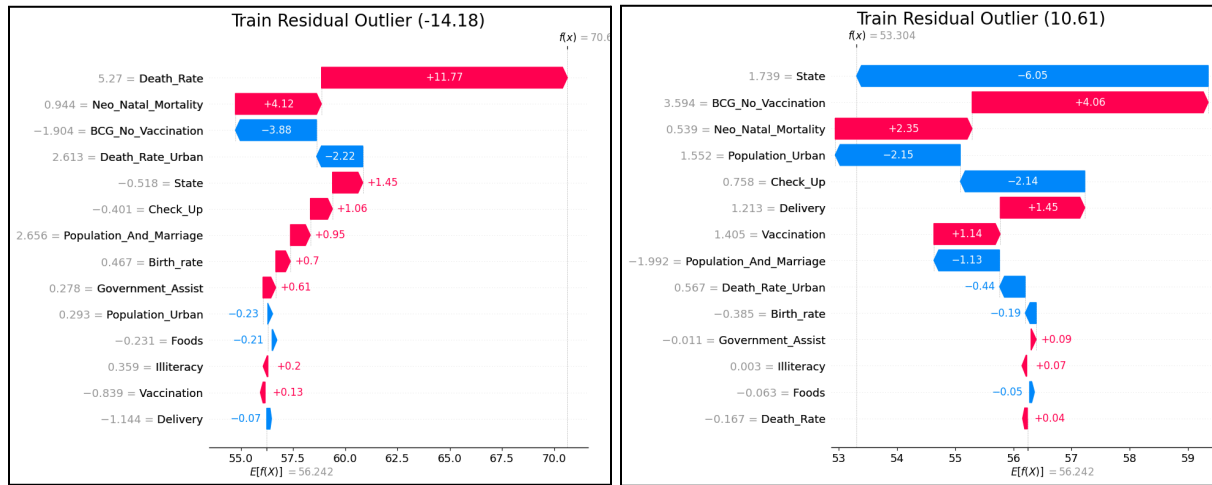


Figure 13. SHAP waterfall plots of an underprediction and an overprediction.

From the overprediction (residual of -14.18) it is clear that the death rate for this district severely inflates the predicted IMR; this implies the presence of instrumental variables that are inflating the general death rate but not IMR directly, which interfered with the model's prediction. The presence of instrumental variables is an extremely important limiting factor for the model because district-specific variables that interfere with the relationship between IMR and the predictors can lead to inaccurate results. The SHAP plot of the underprediction (residual of 10.61) reveals that the district's state is pushing the model's prediction too low. The impact of state on the model's prediction implies that the district is an outlier within its state and an issue may exist at the state-wise level. The potential of a state-level issue is problematic because it may not be properly captured in this districtwise dataset and study.

Conclusion and Future Work

In conclusion, the purpose of this project was to produce a model that could predict districtwise IMR and identify pressing issues in a region's public health. The greatest contributions to the models success were the two stage feature selection process that reduced the featureset from 643 original variables to 14 definable components and the custom ANN architecture that enabled a strong combination of linear and nonlinear learning. Furthermore, a successful deployment app was created to combine the

predictive model with XAI and agentic AI to both produce an IMR prediction and use the prediction's contributing factors to identify district-specific public health problems and recommend potential solutions.

The greatest limitation to this project was the number of observations. Having only 284 severely restricted the learning potential of the models. Furthermore, because the number of rows was far exceeded by the number of features, the vast majority of features could not be used to prevent overfitting. Therefore, future work should emphasize collecting more data. Expanding the collection of data to span more states would not only increase the size of the dataset and quality of the models, but it would also expand the applicable range of the model's deployment.